

BSDS Exercise Set 4 — Detailed Step-by-Step Solutions

Each solution begins with the exact assignment question and the theory needed from the book.

Contents

Question 1

Assignment question.

1. Let $\{S_n : n \geq 0\}$ be the simple symmetric ($p=1/2$) random walk in one dimension with $S_0=0$. (a) Show that $\{|S_n| : n \geq 0\}$ is a Markov chain on $\{0, 1, 2, \dots\}$. Find its transition matrix. (b) Let $M_n = \max_{0 \leq k \leq n} S_k$, and show that $Y_n = M_n - S_n$ defines a Markov chain. Find the transition probabilities of this chain. What happens if the random walk is simple but asymmetric? What happens if the starting position is not the origin, i.e. $S_0 \neq 0$?

Theory used from the book.

A random walk on a graph or on the integers is a Markov chain when the next state depends only on the current state. For a simple random walk, the transition probabilities are determined by the legal one-step moves, and for reflected or transformed processes one checks the Markov property by writing the next value as a function of the present state and one fresh step of randomness. In the book this viewpoint is used repeatedly in the discussion of random walks, gambler's ruin, and reflected barriers.

We treat the two parts separately.

(a) The process $|S_n|$.

For the simple symmetric walk, each increment is ± 1 with probability $1/2$.

- If the current state is 0, then the next step must be to 1, because $|0 \pm 1| = 1$.
- If the current state is $i \geq 1$, then the next absolute value is either $i - 1$ or $i + 1$.

To see this carefully, suppose $|S_n| = i \geq 1$. There are two possibilities for the actual value S_n :

$$S_n = i \quad \text{or} \quad S_n = -i.$$

Because the walk is symmetric, both possibilities lead to the same distribution for $|S_{n+1}|$:

$$\mathbb{P}(|S_{n+1}| = i + 1 \mid |S_n| = i) = \frac{1}{2}, \quad \mathbb{P}(|S_{n+1}| = i - 1 \mid |S_n| = i) = \frac{1}{2}.$$

At $i = 0$ we have

$$\mathbb{P}(|S_{n+1}| = 1 \mid |S_n| = 0) = 1.$$

Hence $\{|S_n|\}$ is a Markov chain on $\{0, 1, 2, \dots\}$ with transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \cdots \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & \cdots \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & \cdots \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

(b) The process $Y_n = M_n - S_n$.

Here $M_n = \max\{S_0, \dots, S_n\}$ and Y_n measures how far the walk is below its current maximum.

Let us check the one-step update.

- If $Y_n = 0$, then $S_n = M_n$ is currently at the maximum.
 - If the next step is $+1$, then the walk makes a new maximum, so $M_{n+1} = M_n + 1$ and $S_{n+1} = S_n + 1$. Therefore

$$Y_{n+1} = M_{n+1} - S_{n+1} = (M_n + 1) - (S_n + 1) = 0.$$

- If the next step is -1 , then $M_{n+1} = M_n$ and $S_{n+1} = S_n - 1$, so

$$Y_{n+1} = M_n - (S_n - 1) = 1.$$

Thus from state 0 we go to 0 with probability $1/2$ and to 1 with probability $1/2$.

- If $Y_n = i \geq 1$, then the walk is strictly below its maximum. In this case a $+1$ step reduces the gap by 1, and a -1 step increases the gap by 1:

$$Y_{n+1} = i - 1 \quad \text{or} \quad i + 1,$$

each with probability $1/2$.

So $\{Y_n\}$ is also a Markov chain on $\{0, 1, 2, \dots\}$ with transition probabilities

$$p_{00} = \frac{1}{2}, \quad p_{01} = \frac{1}{2}, \quad p_{i,i-1} = \frac{1}{2}, \quad p_{i,i+1} = \frac{1}{2} \quad (i \geq 1),$$

and all other probabilities equal to 0.

What changes if the walk is asymmetric?

If the increments are $+1$ with probability p and -1 with probability $q = 1 - p$, then the absolute value process is no longer Markov in general. The reason is that when $|S_n| = i \geq 1$, the next absolute value depends on whether the hidden position is $+i$ or $-i$, and these two cases give different next-step laws unless $p = q = 1/2$.

So:

- for the symmetric walk, $|S_n|$ is Markov;
- for the asymmetric walk, $|S_n|$ is generally not Markov.

If the starting point is not the origin, then for the symmetric walk the same reasoning works: the process $|S_n|$ is still Markov, only the initial state changes from 0 to $|S_0|$. For the asymmetric walk, starting away from the origin does not repair the loss of the Markov property.

Question 2

Assignment question.

2. A die is rolled repeatedly. The results be denoted by $x_i, i=1, \dots$. Which of the following are Markov chains? For those that are, find the transition matrix. (a) The largest number $M_n = \max\{x_1, \dots, x_n\}$ shown up to the n -th roll, i.e. $M_n = \max\{x_1, \dots, x_n\}$. (b) The number N_n of sixes in n rolls, i.e. $N_n = \sum_{k=1}^n 1(x_k = 6)$. (c) The number of throws since the last six, $L_n = n - \max\{k \leq n : x_k = 6\}$, with the convention that if the set is empty, the maximum is taken to be 0. (d) The number of throws until the next six, $R_n = \min\{k \geq n : x_k = 6\} - n$.

Theory used from the book.

A Markov chain is obtained whenever the next value can be expressed as a function of the current value and one fresh die roll. The basic examples in the book include counting successes, waiting times, and record processes built from independent trials.

We use the fact that the die rolls are independent and each face has probability $1/6$.

(a) **Maximum so far.** State space $\{1, \dots, 6\}$ and

$$p_{ij} = \begin{cases} i/6, & j = i, \\ 1/6, & j = i + 1, \dots, 6, \\ 0, & j < i. \end{cases}$$

(b) **Number of sixes.** This is a birth chain:

$$p_{i,i+1} = 1/6, \quad p_{i,i} = 5/6, \quad i \geq 0.$$

(c) **Number of throws since the last six.** This is also Markov:

$$p_{i,0} = 1/6, \quad p_{i,i+1} = 5/6, \quad i \geq 0.$$

(d) **Number of throws until the next six.** This is Markov on $\{1, 2, \dots\}$.

For $i \geq 2$, the remaining time decreases deterministically:

$$p_{i,i-1} = 1.$$

From state 1, the next waiting time is fresh and geometric:

$$p_{1,m} = \left(\frac{5}{6}\right)^{m-1} \frac{1}{6}, \quad m \geq 1.$$

Question 3

Assignment question.

3. Suppose that $x_i, i=1, \dots$ denote the scores obtained at the i -th throw of a fair dice. Set $S_n = (x_{n+1} + x_{n+2}) \bmod 6$, $n \geq 0$. Show that $\{S_n\}$ is a Markov chain. Find the transition probability matrix for this chain.

Theory used from the book.

A standard way to prove a process is Markov is to write the next state as a function of the current state and one fresh independent input. The book also emphasizes that when a process is built from independent observations, one should check whether the present statistic is sufficient for updating the future.

Let $S_n = (\xi_{n+1} + \xi_{n+2}) \bmod 6$ and $S_{n+1} = (\xi_{n+2} + \xi_{n+3}) \bmod 6$. By symmetry, conditioned on $S_n = s$, the variable ξ_{n+2} is uniform on $\{1, \dots, 6\}$. Then one fresh die roll ξ_{n+3} makes the next residue uniform on $\{0, 1, 2, 3, 4, 5\}$. Hence

$$\mathbb{P}(S_{n+1} = t \mid S_n = s) = \frac{1}{6} \quad \forall s, t \in \{0, \dots, 5\}.$$

So the transition matrix is

$$P = \frac{1}{6} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Question 4**Assignment question.**

4. Let $\{X_n\}_{n=0}^\infty$ be a Markov chain on the state space S and transition matrix P . Set, for $k \geq 1$ and $n \geq 0$, $Y_n = (X_n, X_{n+1}, \dots, X_{n+k})$. Show that $\{Y_n\}$ is a Markov chain. What is its transition matrix? The process is often referred to as the snake chain of length $k+1$ associated with $\{X_n\}$.

Theory used from the book.

The book's higher-order transition chapter explains that if one records several consecutive values of a Markov chain, then the enlarged vector process is itself Markov. The next state is obtained by shifting the block one step forward and appending one fresh transition from the last coordinate.

Let $Y_n = (X_n, \dots, X_{n+k})$. Then

$$Y_{n+1} = (X_{n+1}, \dots, X_{n+k+1}).$$

So Y_{n+1} is obtained from Y_n by shifting left and adding one new step from the last coordinate. Therefore, for

$$y = (x_0, \dots, x_k), \quad y' = (x'_0, \dots, x'_k),$$

$$\tilde{p}_{yy'} = \begin{cases} p_{x_k, x'_k}, & x'_0 = x_1, \dots, x'_{k-1} = x_k, \\ 0, & \text{otherwise.} \end{cases}$$

Hence $\{Y_n\}$ is a Markov chain on S^{k+1} .

Question 5

Assignment question.

5. Consider the chain $\{X_n : n \geq 0\}$, where X_n represents the result of independent tosses of a coin with probability of success being p . For $k=2$, the sample space consists of all possible patterns of length 3, i.e. $\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$. Find the transition matrix of this snake chain.

Theory used from the book.

This is the snake chain from the previous question, specialized to length 3. The transition from one length-3 pattern to the next is obtained by dropping the first symbol and appending one fresh coin toss.

Let $q = 1 - p$. Use the state order

$$(HHH, HHT, HTH, HTT, THH, THT, TTH, TTT).$$

The transition matrix is

$$P = \begin{pmatrix} p & q & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & p & q & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & p & q & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & p & q \\ p & q & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & p & q & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & p & q & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & p & q \end{pmatrix}.$$

For example,

$$HHH \rightarrow HHH \text{ with prob. } p, \quad HHH \rightarrow HHT \text{ with prob. } q,$$

and

$$HTH \rightarrow THH \text{ with prob. } p, \quad HTH \rightarrow THT \text{ with prob. } q.$$

Question 6

Assignment question.

6. Let $\{X_n^{(j)} : n \geq 0\}$ be independent sequences of Markov chains for $1 \leq j \leq k$ on the state space $S^{(j)}$ and transition matrix $P^{(j)}$. Set $Y_n = (X_n^{(1)}, X_n^{(2)}, \dots, X_n^{(k)})$. Show that $\{Y_n\}$ is a Markov chain. What is its transition matrix?

Theory used from the book.

The book's product-chain construction says that independent component chains can be combined coordinatewise. The next joint step is determined by the next step of each component, and because the components are independent, the transition probabilities multiply.

Let $Y_n = (X_n^{(1)}, \dots, X_n^{(k)})$. Because the component chains are independent, for states $(x_1, \dots, x_k)$ and $(y_1, \dots, y_k)$,

$$\mathbb{P}(Y_{n+1} = (y_1, \dots, y_k) \mid Y_n = (x_1, \dots, x_k)) = \prod_{j=1}^k p_{x_j y_j}^{(j)}.$$

This depends only on the present vector, so $\{Y_n\}$ is Markov.

Thus the transition matrix is the Kronecker product

$$P^{(Y)} = P_1 \otimes P_2 \otimes \dots \otimes P_k.$$

Question 7

Assignment question.

7. Suppose that a rook and a bishop have been placed at opposite corners of an otherwise vacant chessboard. The pieces move simultaneously and independently of each other to squares where they can legally move. Note that we are allowing the pieces to come to the same square at the same time. Formulate this as a Markov chain. What is its transition matrix?

Theory used from the book.

This is a product chain: the joint next position is obtained by updating each piece independently according to its own legal moves.

Let the state space be the set of ordered pairs

$$S = \{\text{rook squares}\} \times \{\text{bishop squares}\}.$$

A state is (r, b) .

If the rook can move from r to r' in one legal rook move and the bishop can move from b to b' in one legal bishop move, then

$$P((r, b), (r', b')) = P_R(r, r')P_B(b, b').$$

If either move is illegal, the probability is 0.

Hence the full transition matrix is the product-chain matrix

$$P = P_R \otimes P_B.$$

The initial state is the pair of opposite corners.

Question 8

Assignment question.

8. Let $\{X_n\}_{n=0}^\infty$ be a sequence of i.i.d. random variables taking values in $\{-1, 1\}$ with $P(X_n = 1) = p = 1 - P(X_n = -1)$. Let $X_n = x_n x_{n-1} \dots x_1 x_0$, for all $n \geq 0$. Is $\{X_n\}$ a Markov chain?

Theory used from the book.

A process built from overlapping windows of i.i.d. variables may or may not be Markov. The only correct way to decide is to compute the conditional law of the next value given the present value and check whether earlier values matter.

Yes, $\{X_n\}$ is a Markov chain on $\{-1, 1\}$.

Let $q = 1 - p$.

If $X_n = 1$. This means $\xi_n = \xi_{n+1}$. The two possibilities are $(1, 1)$ and $(-1, -1)$. Therefore

$$\mathbb{P}(\xi_{n+1} = 1 \mid X_n = 1) = \frac{p^2}{p^2 + q^2}, \quad \mathbb{P}(\xi_{n+1} = -1 \mid X_n = 1) = \frac{q^2}{p^2 + q^2}.$$

Now

$$\mathbb{P}(X_{n+1} = 1 \mid \xi_{n+1} = 1) = p, \quad \mathbb{P}(X_{n+1} = 1 \mid \xi_{n+1} = -1) = q.$$

Hence

$$\mathbb{P}(X_{n+1} = 1 \mid X_n = 1) = \frac{p^3 + q^3}{p^2 + q^2},$$

and

$$\mathbb{P}(X_{n+1} = -1 \mid X_n = 1) = \frac{pq}{p^2 + q^2}.$$

If $X_n = -1$. Then $\xi_n \neq \xi_{n+1}$. By symmetry,

$$\mathbb{P}(X_{n+1} = 1 \mid X_n = -1) = \frac{1}{2}, \quad \mathbb{P}(X_{n+1} = -1 \mid X_n = -1) = \frac{1}{2}.$$

Therefore, with the state order $(-1, 1)$,

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{pq}{p^2+q^2} & \frac{p^3+q^3}{p^2+q^2} \end{pmatrix}.$$

Question 9

Assignment question.

9. Let $\{x_i\}_{i=1}^n$ be a sequence of i.i.d. integer-valued random variables. Let $S_0=0$ and $S_n=\sum_{i=1}^n x_i$, $n \geq 1$. Also define $X_0=0$, $X_1=x_1$ and for $n \geq 2$, $X_n=x_n+x_{n-1}$, and $Y_0=0$ and for $n \geq 1$, $Y_n=\sum_{k=1}^n S_k$. Which of the following constitute Markov chains: (a) $\{X_n\}$, (b) $\{Y_n\}$, and (c) the sequence of pairs $\{(S_n, Y_n)\}$?

Theory used from the book.

The book repeatedly uses the idea that a process is Markov if the current state contains exactly the information needed to update one step ahead. If some hidden past variable is still needed, then the process is not Markov.

(a) $\{X_n\}$. In general, this is not Markov because

$$X_n = \xi_n + \xi_{n-1}$$

does not determine the hidden variable ξ_n , but the next term

$$X_{n+1} = \xi_{n+1} + \xi_n$$

depends on it.

A concrete counterexample is obtained by taking $\xi_i \in \{0, 1\}$ with probability $1/2$ each. Then:

$$\mathbb{P}(X_{n+1} = 2 \mid X_n = 1, X_{n-1} = 0) = \frac{1}{2},$$

but

$$\mathbb{P}(X_{n+1} = 2 \mid X_n = 1, X_{n-1} = 2) = 0.$$

So $\{X_n\}$ is not Markov.

(b) $\{Y_n\}$. This is also not Markov in general because

$$Y_{n+1} = Y_n + S_{n+1},$$

and $S_{n+1} = S_n + \xi_{n+1}$. The one-dimensional variable Y_n does not contain enough information to reconstruct S_n .

(c) $\{(S_n, Y_n)\}$. This *is* Markov, since

$$S_{n+1} = S_n + \xi_{n+1}, \quad Y_{n+1} = Y_n + S_{n+1} = Y_n + S_n + \xi_{n+1}.$$

Thus the next pair depends only on the current pair and one fresh increment. The transition rule is

$$(s, y) \longrightarrow (s + z, y + s + z)$$

with probability $\mathbb{P}(\xi_1 = z)$. Hence

$$P((s, y), (s', y')) = \begin{cases} \mathbb{P}(\xi_1 = s' - s), & y' = y + s', \\ 0, & \text{otherwise.} \end{cases}$$

Question 10

Assignment question.

10. Find an example of a Markov chain $\{X_n : n \geq 0\}$ on the state space S and $x_0, x_1 \in S$ and $A \subseteq S$ (having at least two elements) such that $P(X_2 = x_2 \mid X_1 = x_1, X_0 = x_0) \neq P(X_2 = x_2 \mid X_1 = x_1)$.

Theory used from the book.

The book warns that conditioning on a set of present states is not the same as conditioning on the exact present state. This is why aggregated or lumped observations may fail to preserve the Markov property unless special structure is present.

Take

$$S = \{0, 1, 2\}$$

and the transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus $0 \rightarrow 1$ deterministically, $1 \rightarrow 0$ deterministically, and $2 \rightarrow 2$ deterministically.

Choose

$$x_0 = 0, \quad x_2 = 0, \quad A_1 = \{1, 2\}.$$

Assume

$$\mathbb{P}(X_0 = 0) = \mathbb{P}(X_0 = 2) = \frac{1}{2}.$$

Then $X_1 \in A_1$ means that X_1 is either 1 or 2, and each occurs with probability $1/2$.

Now,

$$\mathbb{P}(X_2 = 0 \mid X_1 \in A_1, X_0 = 0) = 1,$$

because if $X_0 = 0$ then $X_1 = 1$ surely and then $X_2 = 0$ surely.

But

$$\mathbb{P}(X_2 = 0 \mid X_1 \in A_1) = \frac{1}{2}\mathbb{P}(X_2 = 0 \mid X_1 = 1) + \frac{1}{2}\mathbb{P}(X_2 = 0 \mid X_1 = 2) = \frac{1}{2}(1) + \frac{1}{2}(0) = \frac{1}{2}.$$

Hence the two conditional probabilities are different.

Question 11

Assignment question.

11. Consider the chain defined in Exercise 5. Set $A_0 = \{-HHT, HTT, THT, TTT\}$, $A_1 = \{-HTH, TTH\}$, $A_2 = \{-THH\}$, and $A_3 = \{-HHH\}$. Define $Y_n = i$ if $X_n \in A_i$. Show that the aggregated process is a Markov chain on $\{0, 1, 2, 3\}$. Find its transition matrix.

Theory used from the book.

The book explains that when the state space is partitioned into blocks, an aggregated process may again be Markov if every block has consistent one-step behavior. This is the standard idea behind lumping and higher-order transitions.

Use the snake chain from Question 5 with $q = 1 - p$.

From A_0 , one H sends the process to A_1 and one T sends it back to A_0 :

$$P(0 \rightarrow 1) = p, \quad P(0 \rightarrow 0) = q.$$

From A_1 , one H sends it to A_2 and one T sends it to A_0 :

$$P(1 \rightarrow 2) = p, \quad P(1 \rightarrow 0) = q.$$

From A_2 , one H sends it to A_3 and one T sends it to A_0 :

$$P(2 \rightarrow 3) = p, \quad P(2 \rightarrow 0) = q.$$

From A_3 , one H keeps it in A_3 and one T sends it to A_0 :

$$P(3 \rightarrow 3) = p, \quad P(3 \rightarrow 0) = q.$$

Therefore the aggregated chain is Markov on $\{0, 1, 2, 3\}$ with transition matrix

$$P_Y = \begin{pmatrix} q & p & 0 & 0 \\ q & 0 & p & 0 \\ q & 0 & 0 & p \\ q & 0 & 0 & p \end{pmatrix}.$$

Question 12

Assignment question.

12. Show that not all stochastic matrices can be achieved as P^2 .

Theory used from the book.

A very useful observation in matrix theory is that if a stochastic matrix is a square of another stochastic matrix, then its determinant must be a perfect square of the determinant of the root matrix. This gives a quick obstruction.

If $Q^2 = P$, then

$$\det(P) = \det(Q^2) = (\det Q)^2 \geq 0.$$

So any stochastic matrix with negative determinant cannot be a square of another stochastic matrix.

For example,

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is stochastic, but

$$\det(P) = -1 < 0.$$

Therefore P cannot be written as Q^2 for any stochastic matrix Q . So not every stochastic matrix can occur as P^2 .

Question 13

Assignment question.

13. Set $Z_0=0$ and $Z_n=\max\{X_1, X_2, \dots, X_n\}$, $n \geq 1$, where X_i are i.i.d. random variables taking values in $\{-1, 0, 1, 2, \dots\}$. Show that $\{Z_n : n \geq 0\}$ is a Markov chain. Find $p_{ij}^{(n)}$.

Theory used from the book.

A running maximum is a canonical example of a Markov chain generated from i.i.d. observations. The current maximum contains exactly the information needed to update the next maximum.

Let

$$F(m) = \mathbb{P}(X_1 \leq m), \quad m = 0, 1, 2, \dots,$$

and set $F(-1) = 0$.

If $Z_n = i$, then

$$Z_{n+1} = \max\{i, X_{n+1}\}.$$

So the next value depends only on the current maximum i , not on earlier values. Hence $\{Z_n\}$ is Markov.

Now the n -step transition probabilities are:

$$p_{ij}^{(n)} = 0, \quad j < i.$$

If $j = i$, then all n new samples must be $\leq i$:

$$p_{ii}^{(n)} = F(i)^n.$$

If $j > i$, then the maximum after n steps equals j exactly when all n samples are at most j , but not all are at most $j - 1$:

$$p_{ij}^{(n)} = F(j)^n - F(j-1)^n.$$

Therefore

$$p_{ij}^{(n)} = \begin{cases} 0, & j < i, \\ F(i)^n, & j = i, \\ F(j)^n - F(j-1)^n, & j > i. \end{cases}$$

Question 14

Assignment question.

14. Consider a branching process $\{Z_n\}$. Suppose the offspring distribution is $P(X=0)=0.2$, $P(X=1)=0.5$, $P(X=2)=0.3$. (1) Compute the transition probabilities $p_{ij}^{(n)}$, $p_{i-1,j}^{(n)}$, $p_{i-2,j}^{(n)}$ for $j=0,1,2,3,4$. (2) If the population starts with a single individual ($Z_0=1$), what is the probability that the population is extinct ($Z_1=0$) after one generation? (3) What is the probability that the population has exactly 2 individuals after one generation?

Theory used from the book.

The book's branching-process section states that the next generation is the sum of independent offspring counts, one from each individual in the current generation. Therefore the transition probabilities come from convolution of the offspring law.

Let

$$p_0 = 0.2, \quad p_1 = 0.5, \quad p_2 = 0.3.$$

For a branching process, the next generation size is the sum of the offspring counts of the current individuals.

From state 0.

$$p_{00} = 1, \quad p_{0j} = 0 \text{ for } j \geq 1.$$

From state 1. The next generation has the same distribution as one offspring:

$$(p_{1j})_{j \geq 0} = (0.2, 0.5, 0.3, 0, 0, \dots).$$

From state 2. We square the offspring pgf

$$G(s) = 0.2 + 0.5s + 0.3s^2.$$

Then

$$G(s)^2 = 0.04 + 0.20s + 0.37s^2 + 0.30s^3 + 0.09s^4.$$

So

$$(p_{2j})_{j=0}^4 = (0.04, 0.20, 0.37, 0.30, 0.09).$$

Requested probabilities. Starting from one individual,

$$\mathbb{P}(Z_1 = 0) = p_0 = 0.2,$$

so extinction after one generation has probability 0.2.

Also,

$$\mathbb{P}(Z_1 = 2) = p_2 = 0.3.$$

Summary table.

$$(p_{0j})_{j=0}^4 = (1, 0, 0, 0, 0), \quad (p_{1j})_{j=0}^4 = (0.2, 0.5, 0.3, 0, 0), \quad (p_{2j})_{j=0}^4 = (0.04, 0.20, 0.37, 0.30, 0.09)$$